Deep Semantic Bridging Network for Parallax-Robust Visible-Infrared Multimodal Object Detection in UAV Remote Sensing

Lulu Meng, Xingxing Ju, and Zhi Li

***Abstract*—Visible-infrared (RGB-T) multimodal object detection on uncrewed aerial vehicles (UAVs) is challenged by physical parallax, modality conflicts, and information loss of small targets (sub-32 pixels). Existing fusion methods struggle with precise alignment in dynamic environments and suffer from representational competition, limiting detection accuracy. To address these issues, this article proposes a parallax-aware detection framework, PAO-YOLO, built on a deep semantic bridging architecture. It introduces a modality-invariant feature disentanglement (MIFD) module to decouple geometric structures from modality-specific noise within an orthogonal common subspace. These structural priors then guide an elastic deformation adaptive fusion (EDAF) module, which employs predicted offset fields and deformable convolutions for pixel-level alignment. Additionally, a high-resolution P2 detection layer preserves spatial gradients for enhanced small-target perception. Experiments on the DroneVehicle and DVTOD datasets show that PAO-YOLO achieves 82.2% and 89.2% mAP@0.5, respectively. The method effectively mitigates parallax artifacts and improves sensitivity to small targets, sustaining a robust inference speed of 85.6 frames per second (FPS) on a standard GPU. Code is available at: https://github.com/PAO-YOLO-Team/PAO-YOLO.git**

***Index Terms*—Deep semantic bridging, feature alignment, modality disentanglement, multimodal object detection, uncrewed aerial vehicle (UAV) remote sensing.**

I. Introduction

UNCREWED aerial vehicle (UAV) platforms increasingly utilize visible-infrared (RGB-T) multimodal sensors to ensure robust perception across diverse operational scenarios, including traffic monitoring [1], infrastructure inspection, and disaster relief [2], [3], [4]. Despite their widespread application, single-modality sensors are constrained by inherent physical limitations. Visible-light (VL) sensors provide high-resolution textural details but exhibit significant performance degradation under adverse conditions, such as low illumination, scattering media (e.g., smoke), and intense solar glare [5]. Conversely, thermal infrared (TIR) sensors exploit thermal radiation to detect objects in low-visibility environments [6], yielding infrared (IR) images that are often hindered by limited spatial resolution and a lack of fine-grained textural cues [7], integrating complementary information from heterogeneous sources. However, achieving effective spatial-semantic alignment and representation consistency in dynamic UAV environments [8] remains a significant research challenge, particularly due to the nonrigid parallax and modality distribution shifts encountered in remote sensing. A fundamental constraint in UAV-based perception is the physical parallax arising from the sensor baseline B and focal length f of the electro-optical pod. This physical parallax $\mathbf{d}$ varies inversely with the target distance Z, that is, $d = fB/Z$ [9]. Furthermore, high-frequency mechanical vibrations during flight induce nonrigid deformations that further exacerbate misregistration. Conventional fusion paradigms typically assume strict pixel-level alignment; however, the resulting "ghosting" artifacts and blurred boundaries significantly degrade the precision of boundary box regression. While optical flow-based methods have been explored for compensation, they exhibit limited robustness against high-dynamic nonrigid distortion. Beyond geometric constraints, the inherent modal heterogeneity between VL and TIR presents a significant representation challenge. Traditional

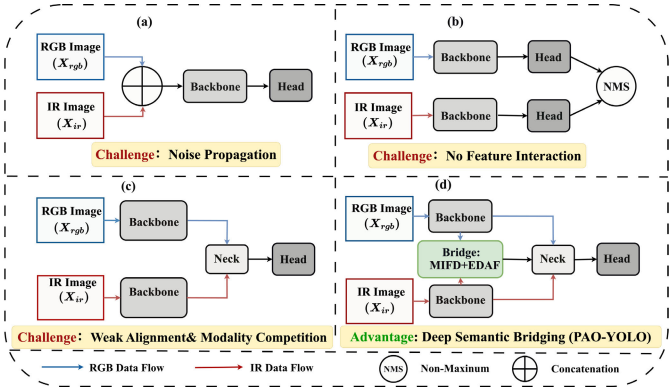


Fig. 1. Schematic of multimodal fusion paradigms. (a) Early data-level stacking. (b) Late decision-level fusion. (c) Conventional feature-level concatenation. (d) PAO-YOLO utilizing DSB for parallax-robust alignment.

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early fusion methods often propagate noise from ineffective modalities, suppressing salient TIR features. Conversely, as illustrated in Fig. 1, late fusion avoids direct gradient conflicts but lacks deep semantic interaction. As noted in [7], when a single modality dominates the representation, it can bias the gradient updates of heterogeneous branches, hindering effective synergy. While recent advances leverage diffusion models for information reconstruction in modality failure scenarios [5], achieving computationally efficient synergy for real-time deployment remains a critical bottleneck. The "Scale Paradox" further limits detection performance in wide fields of view (FoVs), where targets of interest often exhibit a resolution below 32×32 pixels. In conventional architectures like YOLOv8, essential spatial gradients are frequently lost due to successive downsampling operations and low-pass filtering effects [10]. Effective small-target detection requires mechanism-driven feature alignment to preserve fine-grained textural information [11].

The motivation for developing the proposed framework stems from the intrinsic limitations of current state-of-the-art (SOTA) multimodal fusion paradigms when deployed in highly dynamic, unconstrained UAV scenarios. Existing architectures frequently rely on heuristic spatial alignments or unconstrained global attention mechanisms. For instance, recent transformer-based methods like C2Former perform cross-modal interaction globally, which introduces massive computational overhead and ignores the deterministic physical geometry of the sensors. Similarly, spatial alignment methods such as OAFA attempt to learn spatial offsets implicitly but fail to isolate modality-specific noise prior to alignment. This inevitably leads to representational competition, where thermal clutter or optical glare dominates the gradient updates and corrupts the feature fusion process. Motivated by the necessity to fundamentally eliminate this modality interference, we hypothesize that heterogeneous signals must first be decoupled into shared geometric priors and isolated noise before any spatial registration occurs. Compared to existing paradigms, our model provides three distinct advantages.

1) *Mechanism-Driven Alignment:* Rather than heuristic searching, we leverage explicit physical parallax priors to bound the deformation field, achieving robust pixel-level synchronization even under severe UAV vibrations.
2) *Gradient Purification:* By projecting features into an orthogonal subspace, we mathematically suppress modality-specific spectral noise from contaminating the fusion process.
3) *Computational Scalability:* The network circumvents the quadratic complexity of global attention, ensuring real-time inference suitability for edge-computing platforms. To resolve these interconnected challenges, we propose PAO-YOLO, an end-to-end framework centered on the deep semantic bridging (DSB) paradigm. Unlike existing lightweight detectors [12], [13], PAO-YOLO incorporates explicit physical parallax compensation and gradient flow purification. The primary contributions of this article are summarized as follows.
    a) We formalize the PAO-YOLO framework based on the DSB paradigm. By anchoring multimodal interaction between the backbone and neck, the DSB architecture facilitates gradient flow purification and mitigates representational competition prior to multiscale feature aggregation.
    b) We propose a mechanism-driven alignment strategy integrating the modality-invariant feature disentanglement (MIFD) and elastic deformation adaptive fusion (EDAF) modules. Specifically, MIFD maps heterogeneous representations from the feature space into a constrained common subspace $\mathcal{C}$ to isolate modality-invariant structural priors, while EDAF leverages these priors to rectify nonrigid physical parallax through pixel-level elastic registration.
    c) We design a scale-aware detection pipeline that incorporates a high-resolution P2 layer to preserve spatial gradients for sub-32-pixel targets. Experimental results on the DroneVehicle and DVTOD datasets demonstrate that PAO-YOLO establishes a new Pareto frontier, achieving SOTA detection precision while maintaining real-time inference efficiency.

II. Related Work

A. Multimodal Small-Target Detection in UAV Imagery

The detection of sub-32-pixel targets in UAV remote sensing is fundamentally constrained by low signal-to-clutter ratios and high spatial frequency loss [11]. To address localization imprecision, αS-YOLO introduced adaptive angle control coefficients to refine oriented bounding box regression [14]. Subsequent architectures, such as UAV-DETR, optimized processing efficiency by incorporating cascaded linear attention mechanisms [15]. While recent lightweight paradigms like CO-YOLO [12] exploit deformable convolution v4 structures for fruit-picking tasks, they remain insufficient for handling the high-dynamic, nonrigid deformations induced by UAV mechanical vibrations, and rapid ego-motion. The core difficulty in detecting minute targets in UAV imagery lies in preserving the high-frequency spatial components that define subpixel textures. From a signal processing standpoint, successive downsampling in conventional convolutional backbones acts as an aggressive low-pass filter, causing spectral aliasing and irreversible loss of fine-grained geometric gradients [11]. As a result, detecting sub-32-pixel targets becomes a weak-signal problem, where the target's frequency bandwidth is overwhelmed by background clutter. To address this "Scale Paradox," effective architectures must restore local signal saliency by maintaining higher sampling frequencies in shallow layers, capturing critical edge displacements before they are lost to semantic abstraction [10]. Moreover, to address the severe background clutter and feature submergence of small objects in aerial views, recent studies have explored boundary-aware feature fusion and dual-stream attention mechanisms to enhance the geometric saliency of sub-32-pixel targets [16]. Furthermore, to mitigate the extreme target-background imbalance inherent in subpixel perception, enhanced feature learning networks [17] have been recently developed to decouple background noise and adaptively allocate attention to

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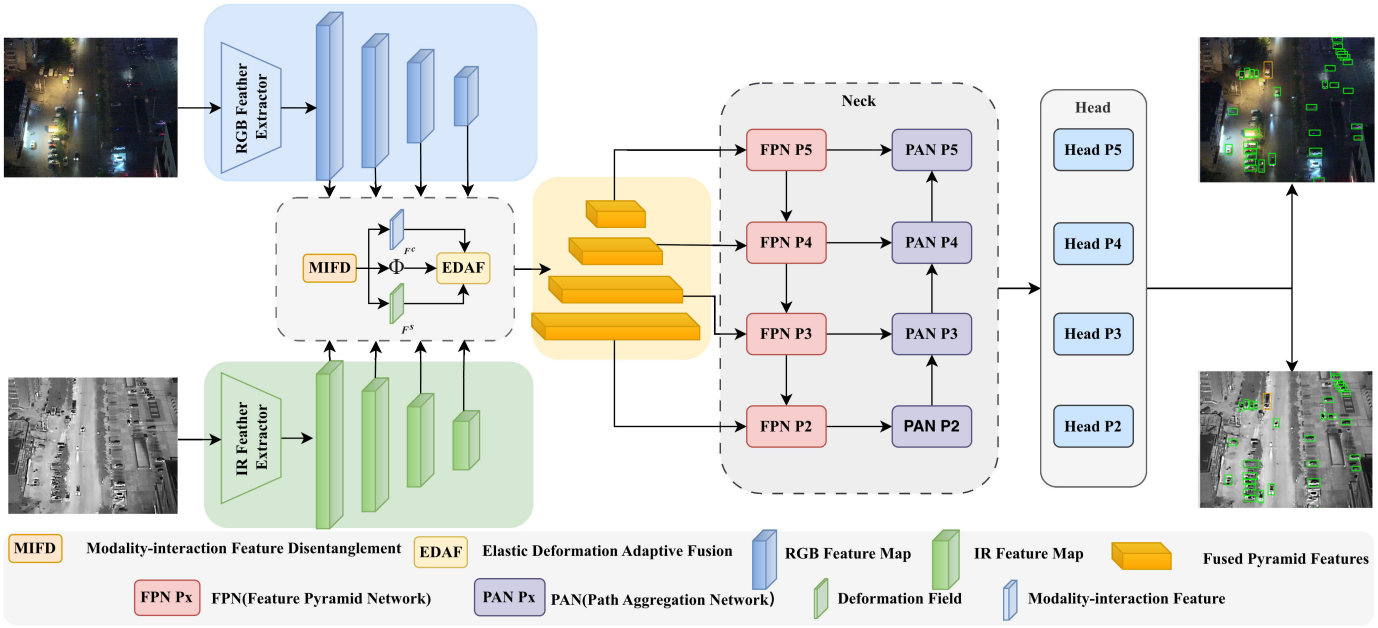


Fig. 2. Schematic of the PAO-YOLO architecture. Multimodal interaction is anchored at the interface between the backbone and the neck to ensure spatial-semantic consistency and gradient flow purification prior to multiscale aggregation.

infrared small targets, preventing target information loss in high-level semantic layers.

B. Heterogeneous Multimodal Fusion Paradigms

Despite employing complex interaction strategies to mitigate distributional discrepancies between VL and TIR signals, current multimodal frameworks remain limited. Whether through MCISFNet's dynamic feature reweighting [3], Remote Det-Mamba's long-range dependency modeling [13], or VTSaRNet's robust feature integration [18], the persistent issues of modality dominance and gradient interference during deep interaction continue to pose significant challenges. To further overcome these challenges, recent SOTA architectures, such as COFNet [19], have demonstrated that establishing contrastive object-aware fusion combined with box-level masking is essential for robust multispectral (RGB-T) object detection, effectively mitigating the adverse effects of modality dominance and background interference.

C. Cross-Modal Alignment and Disentangled Representation

To resolve physical parallax, CoDAF utilizes an offset-guided semantic alignment strategy to rectify cross-modal displacement [20]. CAGTDet further improved consistency via a cascaded alignment-guided transformer [21]. It is worth noting that the pursuit of precise geometric alignment and structural preservation is not unique to 2-D multimodal fusion. In the broader context of spatial perception, similar challenges have driven significant advancements in 3-D object detection, where geometry-guided point generation [22] and adaptive structure-aware cascaded transformers [23] have been effectively utilized to recover incomplete shapes and align complex spatial structures. Inspired by these universal geometric modeling principles, PAO-YOLO shifts from heuristic alignment to a mechanism-driven decoupling strategy for 2-D physical parallax. Building on these developments, PAO-YOLO shifts from heuristic alignment to a mechanism-driven decoupling strategy. By explicitly incorporating the physical parallax prior

$$d = \frac{fB}{Z} \tag{1}$$

where f, B, and Z denote focal length, sensor baseline, and target depth, respectively. Equation (1) characterizes the deterministic geometric relationship between the heterogeneous sensors. By incorporating this physical prior, the proposed framework effectively regularizes the latent offset fields, transitioning the alignment task from a heuristic spatial search to a geometry-constrained registration process. Such physics-embedded regularization has proven effective in complex remote sensing scenarios, where deterministic physical fields (e.g., velocity fields in radar inversion [24]) are utilized to constrain the search space of deep networks, thereby enhancing reconstruction reliability under high-dynamic noise. Specifically, this prior anchors the initial sampling manifold of the deformable kernels, allowing the subsequent EDAF module to prioritize the compensation of high-dynamic nonrigid distortions rather than learning the global sensor displacement from scratch [21]. This physically grounded initialization ensures that the resulting feature-level fusion is executed on a spatially synchronized plane, thereby minimizing the spectral interference induced by physical parallax.

III. Proposed Methodology

A. PAO-YOLO Framework Architecture

Before detailing the internal operations, it is imperative to establish a clear conceptual definition of the core architecture: the DSB Network. In the context of UAV remote sensing, direct feature-level concatenation often fails because raw

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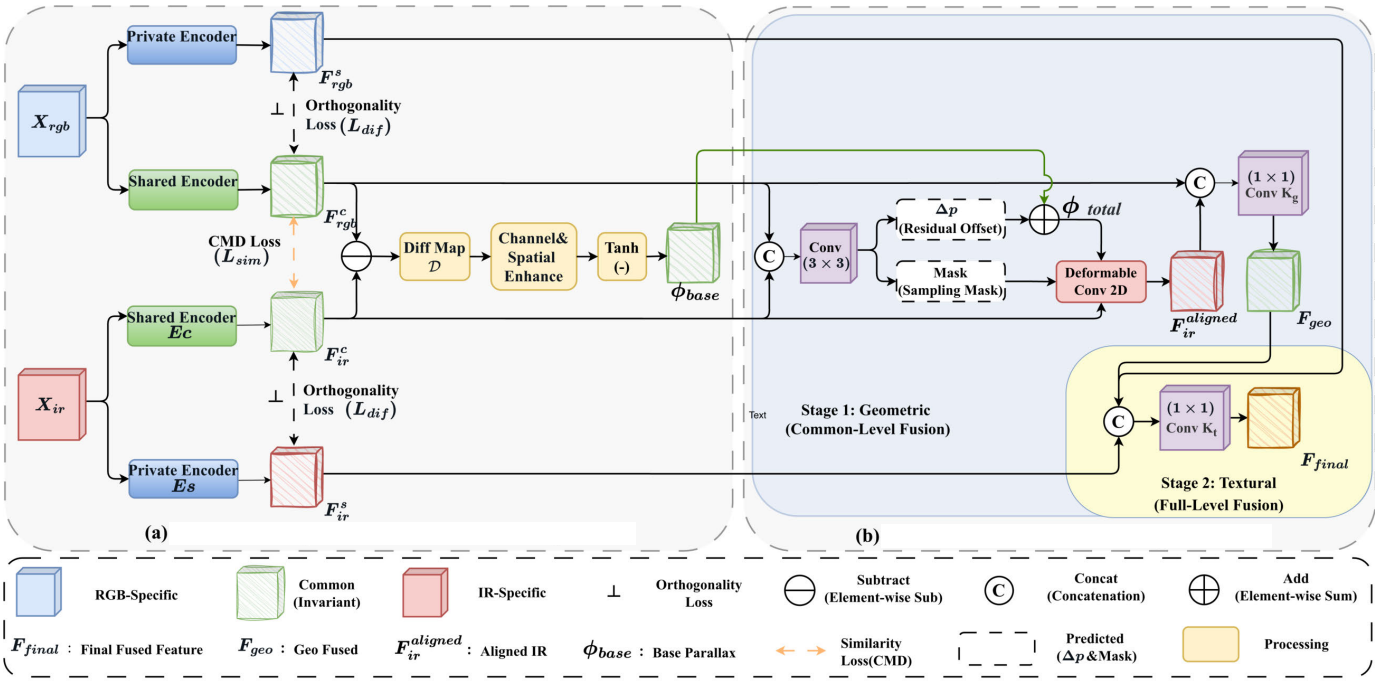


Fig. 3. Detailed architecture of the proposed DSB framework. (a) MIFD module. Heterogeneous features are decoupled into modality-invariant structural priors and modality-specific noise components through orthogonal subspace projection and statistical distribution alignment within a common subspace. (b) EDAF module. By predicting the total offset field ϕ_{total}, the module dynamically guides the convolutional sampling grid toward the target's geometric contours, effectively mitigating parallax and achieving pixel-level spatial alignment.

visible and infrared signals reside in misaligned spatial coordinates and heterogeneous representational distributions. The DSB network is formulated as an intermediate representational conduit strategically interposed at the backbone-neck interface. Its primary function is not simply to aggregate features, but to act as a mathematical "bridge" that maps unaligned heterogeneous tensors into a mutually orthogonal common subspace. Conceptually, the DSB network executes a two-step rectification: it first extracts modality-invariant structural priors (via the MIFD module) to establish a pure geometric consensus, and subsequently utilizes these purified priors to drive an elastic spatial resampling (via the EDAF module). Through this bridging mechanism, the DSB effectively synchronizes the multimodal semantic spaces, guaranteeing that the subsequent feature pyramid operates exclusively on registration-aware and noise-purified representations. To address the inherent physical parallax and gradient interference in UAV-based multimodal perception, we present PAO-YOLO, a framework centered on the DSB paradigm. As illustrated in Fig. 2, the architecture integrates three functional components: a dual-stream backbone for hierarchical feature extraction, the DSB modules (comprising MIFD and EDAF) for parallax-aware alignment, and a set of scale-shared detection heads. Departing from heuristic fusion paradigms, PAO-YOLO conceptualizes multimodal interaction as a DSB operation. This architecture interposes the DSB layer at the backbone-neck interface to execute gradient flow purification. While the backbones function as multiscale signal extractors, the bridging modules (MIFD and EDAF) serve as mechanistic regulators that resolve geometric and distributional discrepancies before the construction of the feature pyramid. Given heterogeneous input tensors $\mathbf{X}_{rgb}$ and $\mathbf{X}_{ir}$, the backbone computes multiscale feature representations $\{\mathbf{F}^i_{rgb}, \mathbf{F}^i_{ir}\}$ for $i \in \{P2, P3, P4, P5\}$. These representations are projected into the DSB layer, where the MIFD module decouples modality-invariant structural priors and estimates the base physical parallax field. Subsequently, the EDAF module implements pixel-level elastic alignment through deformable sampling, effectively rectifying nonrigid distortions. This configuration ensures that the subsequent feature pyramid (Neck) operates on registration-aware and semantically complementary features, thereby suppressing the "ghosting" artifacts prevalent in weakly aligned remote sensing imagery.

B. Modality-Invariant Feature Disentanglement

The representational gap between VL and TIR modalities arises from heterogeneous sensor responses: VL offers texture-rich but noise-prone gradients, while TIR captures thermal data lacking geometric detail. This distributional shift makes direct raw feature correlation prone to physically inconsistent offsets. The MIFD module, as shown in Fig. 3(a), facilitates orthogonal representational disentanglement by projecting heterogeneous tensors from the backbone feature space into a latent common subspace $\mathcal{C}$. This subspace $\mathcal{C}$ is established as the specific output domain for modality-invariant structural descriptors, ensuring they remain distinct from modality-specific spectral noise. The disentanglement process within the common subspace $\mathcal{C}$ is formalized as

$$\mathbf{F}^c_m = E_c(\mathbf{F}_m), \quad \mathbf{F}^s_m = E^m_s(\mathbf{F}_m), \quad m \in \{rgb, ir\} \tag{2}$$

where $\mathbf{F}_m \in \mathbb{R}^{H\times W\times C}$ denotes the multiscale feature tensor extracted from the mth branch of the dual-stream backbone at a specific scale $i \in \{P2, P3, P4, P5\}$, E_c denotes the

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shared-parameter encoder that maps features into the common subspace $\mathcal{C}$, and E_s^m is the private encoder for modality-specific attributes. To facilitate rigorous representational disentanglement, we introduce orthogonality and statistical consistency constraints:

Orthogonality Constraint ($\mathcal{L}_{dif}$): To ensure a disjoint representation of heterogeneous features, the disentanglement process is formulated as a mutual information minimization task between the modality-invariant subspace F_m^c and the modality-specific subspace F_m^s. Since direct minimization of mutual information is computationally intractable in high-dimensional feature spaces, we employ a Frobenius-norm-based orthogonality regularizer as a geometric proxy for decorrelation. This ensures that the two manifolds inhabit orthogonal subspaces within the latent Hilbert space, effectively preventing representational collapse and ensuring that modality-specific spectral noise does not contaminate the invariant geometric priors

$$\mathcal{L}_{dif} = \sum_{m\in\{rgb,ir\}} \left\| \left(F_m^c\right)^\top F_m^s \right\|_F^2 \tag{3}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. This formulation provides a rigorous mechanism for gradient flow separation during joint optimization. The choice of an orthogonality constraint over standard L_1 regularization is motivated by the geometric independence of representational manifolds. While L_1 penalty induces sparsity, it permits the basis vectors of the common and private subspaces to remain collinear, potentially leading to modality noise leakage. In contrast, the Frobenius-norm orthogonality ensures that the modality-specific components are projected into the null space of the geometric priors. Furthermore, we adopt central moment discrepancy (CMD) for statistical alignment as it facilitates explicit order-wise matching of higher order central moments (up to kurtosis). Compared to the kernel-based maximum mean discrepancy (MMD), CMD provides superior sensitivity to the non-Gaussian skews inherent in UAV imagery while maintaining $O(N)$ computational complexity. Theoretically, the disentanglement within the common subspace $\mathcal{C}$ is governed by the information bottleneck (IB) principle. Unlike the attention-based weighting in C2Former, which performs heuristic feature calibration, we formalize the MIFD objective as a dual optimization task: 1) maximizing the mutual information $I(\mathbf{F}_{rgb}^c, \mathbf{F}_{ir}^c)$ to ensure geometric sufficiency; and 2) minimizing the redundancy $I(\mathbf{F}_m^c; \mathbf{F}_m^s)$ through the orthogonality constraint in (3). By projecting heterogeneous signals into the constrained manifold of $\mathcal{C}$, we extract a minimal sufficient representation of the target geometry, effectively filtering modality-specific entropy that would otherwise degrade alignment stability.

Distribution Consistency ($\mathcal{L}_{sim}$): To synchronize the latent distributions of VL and TIR features, we utilize CMD [9], [20] as the statistical distance metric. The mathematical motivation stems from the moment sequence theorem, which posits that probability distributions on a compact interval $[a, b]$ are uniquely determined by their sequence of moments. Unlike kernel-based approaches such as MMD, which operate in an implicit reproducing kernel Hilbert space (RKHS), CMD

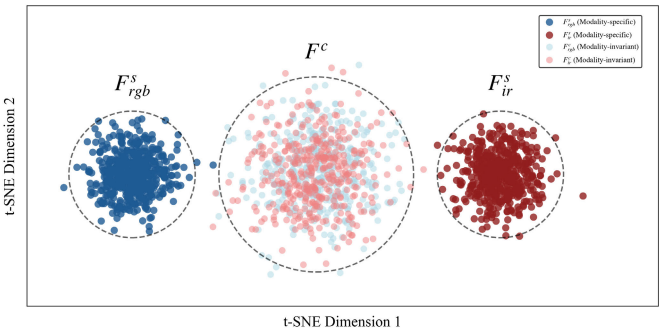


Fig. 4. Distribution analysis of the features extracted by the MIFD module using t-SNE. The high degree of overlap in the central cluster F^C characterizes the successful alignment of modality-invariant features, while the distinct separation of F^s_{rgb} and F^s_{ir} confirms the efficacy of the orthogonality constraint in achieving thorough feature disentanglement.

provides an explicit order-wise matching of higher order central moments

$$CMD(p,q) = \frac{1}{|b-a|}\|E[X] - E\|_2 + \sum_{k=2}^{K} \frac{1}{|b-a|^k}\|c_k(X) - c_k(Y)\|_2 \tag{4}$$

where $c_k(\cdot)$ denotes the kth-order central moment vector. By enforcing consistency across mean, variance, and skewness ($k \in \{1, 2, 3\}$), CMD ensures the convergence of cumulative distribution functions within the feature manifold. This provides a more robust alignment against non-Gaussian sensor noise compared to first-order mean matching, while maintaining a linear computational complexity of $\mathcal{O}(N)$. The base physical parallax ϕ_{base} is subsequently derived from the discrepancy map $\mathbf{D} = \mathbf{F}_{rgb}^c - \mathbf{F}_{ir}^c$, with a *tanh* activation employed to constrain the initial offset within $[-0.5, 0.5]$ for stable guidance [25]. Equation (1) is specifically utilized here to transform the feature-level discrepancy map D into a physically consistent displacement prior ϕ_{base}. By mapping the physical parallax constraints to this base offset field, the MIFD module functions as a geometry-aware alignment engine. This regularizes the initial sampling manifold of the kernels, ensuring that modality-specific attributes are projected into a spatially synchronized common subspace rather than a heuristic latent space [20] To qualitatively assess the effectiveness of the proposed feature disentanglement mechanism, we utilize t-SNE visualization to analyze the distribution of the extracted features in Fig. 4. In the central cluster F^C, a significant overlap between $\mathbf{F}_{rgb}^c$ and $\mathbf{F}_{ir}^c$ is observed, which empirically validates that the consistency constraint $\mathcal{L}_{sim}$ effectively projects heterogeneous signals into a unified modality-invariant latent subspace. Conversely, the modality-specific clusters $\mathbf{F}_{rgb}^s$ and $\mathbf{F}_{ir}^s$ are distinctly separated from each other and from the central common manifold. This clear spatial isolation provides concrete evidence that the orthogonality constraint $\mathcal{L}_{dif}$ successfully suppresses cross-modal interference and prevents information leakage [10]. Consequently, the DSB architecture ensures that subsequent fusion is guided by pure geometric commonalities while preserving independent modality-specific attributes [26].

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C. Elastic Deformation Adaptive Fusion

Nonrigid deformations arising from UAV mechanical vibrations and high-dynamic motion are compensated through the EDAF module, which exploits the base parallax $\boldsymbol{\phi}_{\text{base}}$ as a physical prior for pixel-level elastic correction [27]. To refine the coarse parallax estimate, the module computes a residual offset field $\Delta\mathbf{p}$ derived from the cross-modal common subspace [28]. The aggregate offset field $\boldsymbol{\phi}_{\text{total}}$ is formulated as

$$\phi_{total} = \mathcal{R}\left(\phi_{base}\right) + \Delta p\left(\left[\mathbf{F}_{rgb}^{c}, \mathbf{F}_{ir}^{c}\right]\right) \tag{5}$$

where $[\cdot,\cdot]$ signifies the channel-wise concatenation of features within the common subspace $\mathcal{C}$. $\mathcal{R}(\cdot)$ denotes the spatial tiling operator and $[\cdot,\cdot]$ signifies feature concatenation. The infrared feature map is subsequently resampled using $\boldsymbol{\phi}_{\text{total}}$ to produce the aligned representation $\mathbf{F}_{\text{ir}}^{\text{aligned}}$ via a deformable convolution kernel [29]. This physically grounded initialization serves as a geometry-aware anchor for the deformable kernels. Unlike purely data-driven offsets, which navigate a maximum-entropy search space, the deterministic prior ϕ_{base} regularizes the displacement manifold to a low-entropy state. This aligns with the advanced paradigm of physics-informed neural networks in geoscience, which leverages hard-coded physical laws to prevent models from converging to nonphysical local optima in texture-sparse regions [24]. By transforming the registration task from an unconstrained spatial search into a local residual-gradient descent process, the EDAF module prioritizes the compensation of high-dynamic nonrigid perturbations, thereby enhancing localization invariance under extreme mechanical vibrations. Departing from the unconstrained spatial search employed in CoDAF, the aggregate offset field ϕ_{total} in EDAF is reframed as a geometry-constrained variational optimization problem. By incorporating the deterministic physical parallax prior ϕ_{base} from (1), we effectively restrict the sampling manifold to a low-entropy state. This physically grounded initialization transforms the alignment from a stochastic search into a local residual-gradient descent task, maximizing the conditional entropy $H(\text{alignment}|\phi_{base})$. Consequently, the model achieves superior convergence by prioritizing the rectification of high-dynamic nonrigid perturbations rather than learning the global sensor displacement from scratch. The learned total offset fields ϕ_{total} are visualized in Fig. 3(b). In contrast to standard convolution with fixed-grid sampling, the predictive residual offset Δp facilitates adaptive reconfiguration of the receptive fields based on local feature distributions. As illustrated in the vector field map, the sampling trajectories (represented by blue arrows) converge from the initial regular grid toward the structural boundaries of targets in the infrared modality. This spatial resampling mechanism effectively compensates for nonrigid deformations induced by physical parallax and UAV platform vibrations [21], [30]. By ensuring that cross-modal fusion is executed on spatially synchronized feature planes, the EDAF module suppresses the ghosting artifacts typically associated with weakly aligned multimodal signals [10], [31]. To maximize cross-modal synergy, a two-stage fusion strategy is employed, sequentially integrating geometric and textural components. The geometric fusion is defined as follows:

$$\mathbf{F}_{\text{geo}} = \sigma\left(\left[\mathbf{F}_{\text{rgb}}^{c}, \mathbf{F}_{\text{ir}}^{\text{aligned}}\right] * \mathbf{K}_{g}\right) \tag{6}$$

where $\sigma(\cdot)$ signifies the nonlinear activation regularizing the fused response, and $[\cdot,\cdot]$ denotes the channel-wise concatenation operator. The term $\mathbf{F}_{ir}^{aligned} \in \mathbb{R}^{H\times W\times C}$ represents the infrared feature manifold spatially resampled via the aggregate offset field ϕ_{total} to ensure pixel-level synchronization, while $\mathbf{K}_g$ is the learnable kernel parameterized to capture cross-modal geometric dependencies

$$\mathbf{F}_{\text{fused}} = \sigma\left(\left[\mathbf{F}_{\text{geo}}, \mathbf{F}_{\text{rgb}}^{s}, \mathbf{F}_{\text{ir}}^{s}\right] * \mathbf{K}_{\text{t}}\right) \tag{7}$$

where $\mathbf{K}_t$ represents the textural fusion kernel optimized to integrate modality-specific signatures $\mathbf{F}_m^s$ while mitigating modality competition during gradient back propagation. The two-stage fusion architecture in (6) and (7) implements a hierarchical integration strategy that preserves the structural integrity of sub-32-pixel targets. Initializing fusion with purified geometric descriptors from subspace $\mathcal{C}$ prevents modality dominance by stabilizing the geometric manifold before textural attributes are integrated. This sequential aggregation maximizes cross-modal semantic complementarity during back propagation.

D. Scale-Aware Feature Preservation

The "Scale Paradox" in wide-field UAV imagery stems from targets exhibiting resolutions below 32×32 pixels [11]. To preserve fine-grained textural gradients essential for detecting minute objects, PAO-YOLO incorporates a high-resolution P2 detection layer with a stride of $s = 4$ [10]. We cascade the MIFD and EDAF modules at this finest spatial scale to maximize the signal-to-clutter ratio [9]. By maintaining a higher sampling frequency, the P2 layer facilitates the capture of minute textural edge displacements. This architectural configuration effectively resolves information loss for distant vehicles that typically vanish during successive downsampling and low-pass filtering in deeper convolutional stages [10].

E. Loss Function

To supervise the feature disentanglement process and optimize detection precision, the multiobjective loss function is formalized as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \lambda_1\mathcal{L}_{\text{sim}} + \lambda_2\mathcal{L}_{\text{dif}} + \lambda_3\mathcal{L}_{\text{sem}} \tag{8}$$

where $\mathcal{L}_{\text{det}}$ represents the task-specific detection loss formulated in [28]. The distribution consistency term $\mathcal{L}_{sim}$ employs CMD to enforce statistical alignment between modality-invariant distributions specifically within the MIFD-defined common subspace $\mathcal{C}$ [9]. This ensures that features are purified in the output manifold before multiscale fusion. To ensure representational independence, $\mathcal{L}_{\text{dif}}$ as defined in (3). While $\mathcal{L}_{\text{sem}}$ signifies the semantic consistency loss [10]. The balancing coefficients λ_1, λ_2, and λ_3 are introduced to harmonize the multiobjective optimization landscape, ensuring that the auxiliary regularizations facilitate gradient purification without overwhelming the primary detection objective.

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In our implementation, these hyperparameters are empirically constrained within the interval $[0.01, 0.1]$ via grid search to maintain alignment losses at approximately 10% of the total magnitude. This composite loss function design ensures that PAO-YOLO converges toward a physically consistent state by suppressing modality competition during deep interaction, thereby significantly enhancing detection invariance under nonrigid deformations and physical parallax [30]. To ensure representational independence, $\mathcal{L}_{\text{dif}}$ is defined in (3). This minimizes the correlation between the invariant and modality-specific manifolds to prevent information leakage [9]. Finally, the semantic consistency loss $\mathcal{L}_{\text{sem}}$ utilizes the fused representation as a distillation-based teacher signal to regularize the decoupled features during training [9].

Summary Comparison and Systematic Advantages: To further elucidate the superiority of the proposed PAO-YOLO framework, we provide a systematic synthesis against established paradigms such as OAFA [31], RGFNet [32], and C2Former [20] across three critical dimensions.

1) *Physical Parallax Modeling:* Unlike heuristic alignment strategies in OAFA [31] that rely purely on appearance-based offsets, or the unconstrained global attention in C2Former [20], which lacks geometric grounding, PAO-YOLO formalizes the cross-modal registration as a mechanism-driven process. By mapping the physical parallax prior $d = fB/Z$ into the MIFD-EDAF framework, our model achieves robust pixel-level elastic alignment, effectively suppressing nonrigid artifacts induced by high-dynamic perturbations.
2) *Gradient Flow Purification:* While RGFNet [32] utilizes reflectance-based guidance, which often propagates photometric noise under extreme illumination, our DSB architecture implements an orthogonality constraint via the Frobenius norm. This facilitates the disentanglement of modality-invariant structural gradients from modality-specific spectral noise, thereby mitigating the "modality dominance" trap and ensuring purified gradient propagation during joint optimization.
3) *Small Target Preservation:* In contrast to C2Former [20] and its variants, which suffer from catastrophic spatial frequency loss during aggressive downsampling or heavy attention operations, PAO-YOLO anchors its interaction pipeline at a high-resolution P2 layer ($s = 4$). This configuration maximizes the signal-to-clutter ratio for sub-32-pixel instances, preserving critical high-frequency textural cues that are typically filtered out in deeper convolutional stages.

IV. Experiments

A. Datasets and Evaluation Metrics

The proposed framework is evaluated on the DroneVehicle and DVTOD datasets, which offer complementary signal characteristics for UAV-based sensing. The DroneVehicle dataset [33] provides 28 439 VL-TIR pairs with 953 087 instances, primarily posing challenges related to high spatial frequency loss in sub-32-pixel targets and spectral distribution shifts during day–night transitions. The DVTOD dataset [30] specifically focuses on nonrigid misregistration caused by sensor parallax, featuring 2179 pairs with explicit spatial biases. The inclusion of diverse meteorological conditions (e.g., fog and precipitation) introduces nonlinear signal attenuation, providing a rigorous benchmark for testing the robustness of the model's elastic alignment.

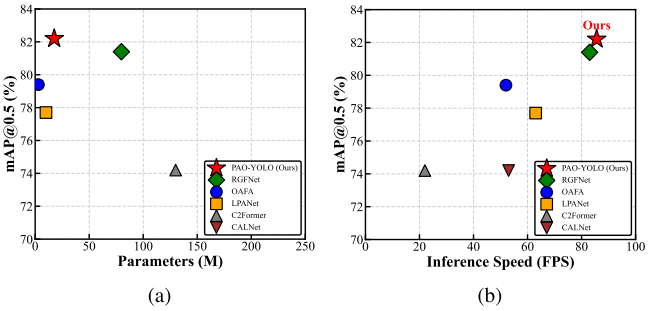


Fig. 5. Performance tradeoffs on the DroneVehicle dataset. (a) mAP versus number of parameters. (b) mAP versus inference speed (FPS).

B. Implementation Details

The proposed PAO-YOLO framework is implemented using the PyTorch deep learning library. All experiments are executed on a workstation equipped with an Intel Core CPU and a single NVIDIA RTX 4090 GPU with 24 GB of VRAM. During the training phase, all input multimodal image pairs are resized to a uniform spatial resolution of 640×640 pixels. The network is optimized end-to-end for 300 epochs utilizing the AdamW optimizer, with a weight decay of 0.05 and a momentum parameter of 0.937. To maximize GPU utilization while preventing memory overflow during the high-resolution feature alignment, the batch size is set to 16. The initial learning rate is set to 1×10^{-3}, which is subsequently decayed to a minimum of 1×10^{-5} following a cosine annealing schedule after a three-epoch linear warmup. To prevent overfitting and improve model generalization in complex UAV sensing scenarios, a standard data augmentation pipeline is applied, encompassing random affine transformations, HSV color-space perturbations, and Mosaic augmentation. Notably, to ensure the stability of the EDAF module's spatial alignment, strong augmentations like Mosaic are disabled during the final 15 epochs. During the inference phase, the intersection over union (IoU) threshold for nonmaximum suppression (NMS) is set to 0.65. To ensure a rigorous evaluation of the real-time performance, the inference speed [frames per second (FPS)] is strictly measured on the aforementioned RTX 4090 GPU with the batch size set to 1. This configuration specifically emulates the sequential, frame-by-frame data stream processing characteristic of actual UAV flight operations. The reported FPS explicitly encompasses the entire algorithmic pipeline, including data loading, network forward propagation, and NMS postprocessing.

C. Performance Benchmarking and SOTA Comparison

To rigorously assess the practical utility of the proposed network in resource-constrained UAV environments, we examine the tradeoff between detection accuracy and computational

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TABLE I

QUANTITATIVE COMPARISON WITH SOTA METHODS ON THE DRONEVEHICLE DATASET. RESULTS ARE REPORTED IN MAP@0.5 (%) AND INFERENCE SPEED (FPS)

Method	Backbone	Type	Modality	Car	Truck	F-Car	Bus	Van	mAP	FPS
DroneVehicle val.										
Faster R-CNN (OBB) [34]	ResNet50	Two-Stage	VL	97.70	42.00	34.00	76.90	37.70	54.10	21.50
YOLOv5 (OBB) [35]	CSPDarkNet-S	One-Stage		89.20	59.70	43.00	78.80	43.90	62.90	145.30
YOLOv8 (OBB) [26]	CSPDarkNet-S	One-Stage		93.40	70.60	56.60	92.10	59.80	74.50	180.50
RetinaNet (OBB) [36]	ResNet50	One-Stage	IR	88.80	35.40	39.50	76.50	32.10	54.50	61.20
Faster R-CNN (OBB) [34]	ResNet50	Two-Stage		89.70	41.00	43.10	86.30	41.20	60.30	21.50
YOLOv5 (OBB) [35]	CSPDarkNet-S	One-Stage		89.50	59.90	49.50	86.70	45.10	66.10	145.30
YOLOv8 (OBB) [26]	CSPDarkNet-S	One-Stage		97.30	73.10	67.80	93.60	61.20	78.60	180.50
Halfway Fusion [37]	ResNet50	Two-Stage	VL+IR	89.90	60.30	55.50	89.00	46.30	68.20	80.00
CIAN (OBB) [38]	ResNet50	One-Stage		90.00	62.50	60.20	88.90	49.60	70.20	84.50
OAFA [31]	CSPDarkNet-S	Two-Stage		90.30	78.20	75.70	90.10	64.90	79.80	52.00
DPDERT [39]	ResNet50	One-Stage		90.30	78.20	75.70	90.10	67.50	81.00	-
LPANet [39]	CSPDarkNet-S	Two-Stage		90.40	80.60	**76.30**	90.10	64.90	79.80	63.00
ADMPF [40]	CSPDarkNet-S	One-Stage		**97.90**	**82.80**	69.30	95.90	64.80	82.10	-
Baseline	CSPDarkNet-S	One-Stage		97.60	77.00	69.30	91.60	67.30	80.60	**245.60**
PAO-YOLO (Ours)	**CSPDarkNet-S**	**One-Stage**		96.80	81.00	75.70	**96.80**	**74.40**	**84.10**	85.60
DroneVehicle test.										
RetinaNet (OBB) [36]	ResNet50	One-Stage	VL	67.50	28.20	13.70	62.10	19.30	38.20	61.20
ReDet [26]	ResNet50	One-Stage		69.50	47.90	31.50	77.40	29.00	51.10	-
Gliding Vertex [37]	ResNet50	Two-Stage		75.80	46.10	33.80	68.10	38.70	52.50	-
YOLOv5 (OBB) [35]	CSPDarkNet-S	One-Stage		76.20	48.90	35.50	68.90	40.40	54.00	145.30
YOLOv8 (OBB) [26]	CSPDarkNet-S	One-Stage		92.30	69.00	54.70	92.10	56.00	72.80	180.50
RetinaNet (OBB) [36]	ResNet50	One-Stage	IR	79.90	32.80	28.00	67.30	16.40	44.90	15.0
ReDet [26]	ResNet50	One-Stage		89.50	54.00	42.80	79.90	36.60	60.60	-
YOLOv5 (OBB) [35]	CSPDarkNet-S	One-Stage		97.30	74.80	61.70	90.90	56.40	76.20	145.30
YOLOv8 (OBB) [26]	CSPDarkNet-S	One-Stage		97.30	73.10	67.80	93.60	61.20	78.60	180.50
UA-CMDet [33]	ResNet50	Two-Stage	VL+IR	87.50	60.70	46.80	87.10	38.00	64.00	-
CALNet [41]	CSPDarkNet-S	One-Stage		90.30	76.20	63.00	89.10	58.50	74.20	53.00
C2Former [20]	ResNet50	One-Stage		90.20	68.30	64.40	89.80	58.50	74.20	22.50
LPANet [39]	CSPDarkNet-S	Two-Stage		90.40	78.00	65.00	89.50	65.40	77.70	63.00
OAFA [31]	CSPDarkNet-S	Alignment		90.30	76.80	**73.30**	90.30	66.00	79.40	52.00
RGFNet [32]	CSPDarkNet-S	One-Stage		98.40	81.10	68.70	95.8	63.00	81.40	83.00
Baseline	CSPDarkNet-S	One-Stage		97.30	74.80	61.70	90.90	56.40	78.90	**245.60**
PAO-YOLO (Ours)	**CSPDarkNet-S**	**One-Stage**		**98.60**	**82.50**	70.80	**97.10**	**71.50**	**82.20**	85.60

overhead. Fig. 5(a) benchmarks PAO-YOLO against representative SOTA methods on the DroneVehicle dataset [2], [3].

As delineated in Fig. 5(a), PAO-YOLO achieves the highest detection fidelity while maintaining a significantly smaller footprint, effectively defining the upper left boundary of the accuracy-parameter space. Specifically, PAO-YOLO yields an 8.0% mAP improvement over the transformer-based C2Former [5] while utilizing merely 13.6% of its parameter count. This performance envelope demonstrates that the DSB architecture enables efficient cross-modal interaction through mechanism-driven rectification rather than simple network scaling [21]. Furthermore, the real-time processing capability is evaluated in Fig. 5(b). Although recent methods such as OAFA [31] and CALNet demonstrate competitive performance, their inference speeds (52 FPS and 53 FPS, respectively) are restricted by complex alignment overheads. While RGFNet [32] achieves a competitive speed of 83 FPS, it falls short in detection fidelity (81.4%). In contrast, PAO-YOLO maintains a superior throughput of 85.6 FPS with the highest accuracy (82.2%), successfully establishing a new Pareto frontier. These results underscore that the proposed parallax-aware offset learning effectively mitigates geometric noise while preserving a low-latency signal processing pipeline, which is paramount for mission-critical UAV sensing. To quantitatively assess the

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TABLE II
MECHANISTIC COMPARISON OF ALTERNATIVE DESIGN CHOICES. ALL RESULTS ARE REPORTED IN MAP@0.5 (%). † DENOTES THE PROPOSED PAO-YOLO CONFIGURATION

Design Dimension	Mechanism	Strategy	Goal	mAP (%)
Disentanglement	MIFD	L_1 Regularization	Sparsity	80.20
	MIFD	Orthogonality†	Independence	**81.40**
Statistical Alignment	MIFD	MMD (RKHS)	Kernel-based	80.70
	MIFD	CMD†	Moment-based	**81.40**
Geometric Alignment	EDAF	Heuristic DCN	Black-box	81.30
	EDAF	Physics-Prior†	Physics-informed	**82.20**

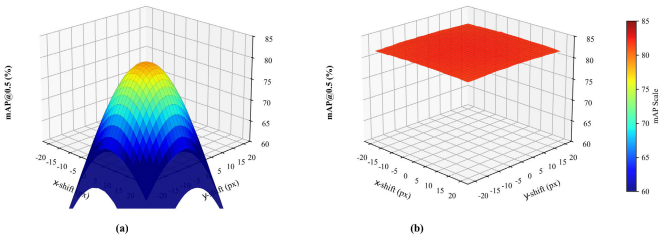


Fig. 6. Quantitative robustness analysis against simulated spatial misalignment. (a) Baseline model suffers from significant performance degradation due to geometric synchronization failure. (b) Our PAO-YOLO achieves stable detection accuracy across large offsets, validating the efficacy of the EDAF module in rectifying physical parallax via dynamic offset learning [26], [30].

misalignment tolerance of the proposed architecture, we conducted a position shift experiment following the protocol in [32]. Fig. 6 characterizes the mAP@0.5 performance surface as a function of simulated spatial offsets within a grid of $\{(\Delta x, \Delta y) \mid \Delta x, \Delta y \in [-20, 20]\}$ pixels. As illustrated in Fig. 6, the baseline model exhibits a sharp parabolic performance decay, where detection accuracy drops from 78.9% to below 65% as misalignment increases, indicating extreme sensitivity to spatial phase shifts. In contrast, PAO-YOLO maintains a high-performance plateau across the entire range, demonstrating remarkable performance invariance. This stability provides empirical validation that the EDAF module effectively compensates for physical parallax by dynamically reconfiguring the convolution sampling trajectories through the learned offset fields ϕ_{total}. We benchmark PAO-YOLO against representative single-modality and multimodal detection paradigms. The comparative results on the DroneVehicle dataset are summarized in Table I. Experimental validation indicates that PAO-YOLO achieves an mAP@0.5 of 82.2% on the DroneVehicle test set, yielding quantifiable gains over contemporary frameworks such as OAFA [31] and RGFNet [32]. This improvement is attributed to the synergistic effect of physical parallax compensation and gradient flow purification enabled by the semantic bridging architecture.

D. Mechanistic Investigation

To substantiate the underlying mechanisms of PAO-YOLO and elucidate the theoretical necessity of our architectural choices, we conduct a comparative study against alternative design paradigms II.

1) Independence Versus Sparsity for Feature Purification: As shown in Table II, substituting the orthogonality constraint with L_1 regularization results in a 1.2% mAP decay. This drop confirms that mere sparse selection is insufficient to block cross-modal interference; explicit perpendicularity in the representation space is required to ensure that gradient updates for geometric alignment are not contaminated by spectral-specific artifacts.

2) CMD Versus MMD for Distributional Robustness: Replacing CMD with MMD leads to a 0.7% precision loss. This performance gap highlights MMD's sensitivity to kernel bandwidth σ, which often fails to capture the nonstationary distributions encountered during UAV altitude changes. CMD's granular characterization of the second, third, and fourth central moments ensures a more robust semantic consensus between heterogeneous sensors.

3) Physics-Prior Versus Heuristic Search: The 0.9% gain achieved by our physics-informed initialization over a heuristic DCN validates the necessity of geometric anchoring. Without the physical parallax prior, deformable kernels often converge to local minima in texture-sparse regions, whereas our approach maintains localization stability by focusing exclusively on nonrigid residual refinement.

E. Ablation Study

To systematically evaluate the individual contributions and computational overhead of each proposed component in PAO-YOLO, we conduct a comprehensive ablation study on the DroneVehicle dataset. In UAV-based remote sensing, deploying models on resource-constrained edge devices requires a rigorous tradeoff between detection precision and computational cost. In response to the strict size, weight, and power (SWaP) constraints, we expand our evaluation beyond accuracy to report the number of parameters (Params), floating-point operations (GFLOPs), and real-time inference speed (FPS) alongside the mAP metric. The baseline is established using a dual-stream YOLOv8s architecture with element-wise addition for late fusion. The inference speed is measured on a single NVIDIA RTX 4090 GPU with a batch size of 1 at a 640 × 640 resolution. The quantitative trajectory of the ablation is detailed in Table III.

1) Ablation Study on Proposed Modules: To systematically evaluate the independent contributions, synergistic effects, and

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TABLE III

ABLATION STUDY OF THE PROPOSED MODULES ON THE DRONEVEHICLE TEST SET. THE BASELINE IS A DUAL-STREAM YOLOV8S. FPS IS MEASURED ON A SINGLE NVIDIA RTX 4090 GPU

P2	MIFD	EDAF	Backbone	mAP@0.5 (%)	Params (M)	GFLOPs	FPS
			CSPDarkNet-S	78.90	16.50	43.50	245.60
✓			CSPDarkNet-S	79.60	16.92	58.20	155.30
	✓		CSPDarkNet-S	79.80	16.85	45.40	218.10
		✓	CSPDarkNet-S	80.10	16.90	45.80	168.20
✓	✓		CSPDarkNet-S	80.70	17.27	60.10	138.20
✓		✓	CSPDarkNet-S	81.10	17.32	60.50	105.10
	✓	✓	CSPDarkNet-S	80.90	17.25	47.70	142.00
✓	✓	✓	**CSPDarkNet-S**	**82.20**	**17.65**	**62.40**	**85.60**

TABLE IV

QUANTITATIVE COMPARISON OF DETECTION PERFORMANCE ON THE DVTOD DATASET. RESULTS ARE REPORTED IN MAP@0.5 (%)

Method	Backbone	Type	Modality	Person	Car	Bicycle	mAP@0.5
YOLOv5 [33]	CSPDarkNet-S	One-Stage	VL	23.30	50.20	26.70	33.40
YOLOv8 [32]	CSPDarkNet-S	One-Stage		28.40	52.70	24.70	35.30
YOLOv10[42]	CSPDarkNet-S	One-Stage		30.40	50.80	23.80	35.00
YOLOv5 [33]	CSPDarkNet-S	One-Stage	IR	86.20	75.90	86.20	82.80
YOLOv8[32]	CSPDarkNet-S	One-Stage		86.20	75.60	88.10	83.30
YOLOv10 [42]	CSPDarkNet-S	One-Stage		85.90	74.90	85.30	82.10
YOLOv5+Add	CSPDarkNet-S	One-Stage	VL+IR	88.80	74.30	74.60	79.20
CET[43]	CSPDarkNet-S	One-Stage		88.90	79.00	81.30	82.70
CMA-Det [44]	CSPDarkNet-S	One-Stage		90.30	81.60	83.10	85.00
RGFNet [32]	CSPDarkNet-S	One-Stage		92.00	84.60	89.10	88.50
Baseline	CSPDarkNet-S	One-Stage		91.60	82.10	85.80	86.50
PAO-YOLO (Ours)	**CSPDarkNet-S**	**One-Stage**		**92.60**	**85.20**	**89.80**	**89.20**

computational overhead of the proposed modules, we conduct a fully orthogonal ablation study on the DroneVehicle dataset. The baseline is a dual-stream CSPDarkNet-S detector utilizing element-wise addition for feature fusion. To directly address the deployment constraints in UAV sensing, we explicitly report the parameter count (Params), theoretical computation (GFLOPs), and empirical inference speed (FPS). The FPS is strictly evaluated on a single NVIDIA RTX 4090 GPU with a batch size of 1. The quantitative results are detailed in Table III.

a) High-Resolution P2 layer: Detecting sub-32-pixel targets in UAV imagery requires preserving high-frequency spatial gradients. As shown in Table III, introducing the P2 layer (stride of 4) to the baseline improves the mAP@0.5 from 78.90% to 79.60%. However, processing larger feature maps inherently increases the memory bandwidth footprint, raising the computational load from 43.50 to 58.20 GFLOPs and dropping the frame rate from 245.60 to 155.30 FPS.

b) Independent Contributions of MIFD and EDAF: When applied independently, both the MIFD and the EDAF modules exhibit steady performance gains, yielding 79.80% (+0.9%) and 80.10% (+1.2%) mAP@0.5, respectively. MIFD introduces a marginal 0.35 M parameters and 1.90 GFLOPs, sustaining a high inference speed of 218.10 FPS. This efficiency is achieved because the computationally intensive orthogonality constraints are strictly confined to backward propagation; during inference, MIFD simplifies into a dense linear projection. Conversely, EDAF adds minimal parameters (+0.40 M) and computation (+2.30 GFLOPs) but causes a more noticeable FPS drop (245.60 $\rightarrow$ 168.20 FPS) due to the scattered memory read patterns inherent to deformable convolutions.

c) Synergistic Effect and Complexity Tradeoffs: The orthogonal results reveal a strong synergistic coupling among the proposed modules when integrated into the full architecture. The full PAO-YOLO model integrates all three components, peaking at 82.20% mAP@0.5. This overall +3.3% accuracy gain (82.20% versus 78.90%) noticeably exceeds the arithmetic sum of their individual improvements (+2.8%, derived from 0.7% +0.9% +1.2%). It empirically demonstrates that filtering modality-specific noise via MIFD provides a purified geometric prior, which successfully stabilizes the deformation field in EDAF and prevents semantic noise from contaminating the spatial resampling process across high-resolution scales. Regarding computational complexity, the full PAO-YOLO requires 17.65 M parameters and 62.40 GFLOPs. Although the dispersed sampling pattern of EDAF and the high-resolution feature maps of P2 inevitably disrupt the GPU L2 cache locality, constraining the deformation field with the physical parallax prior prevents severe

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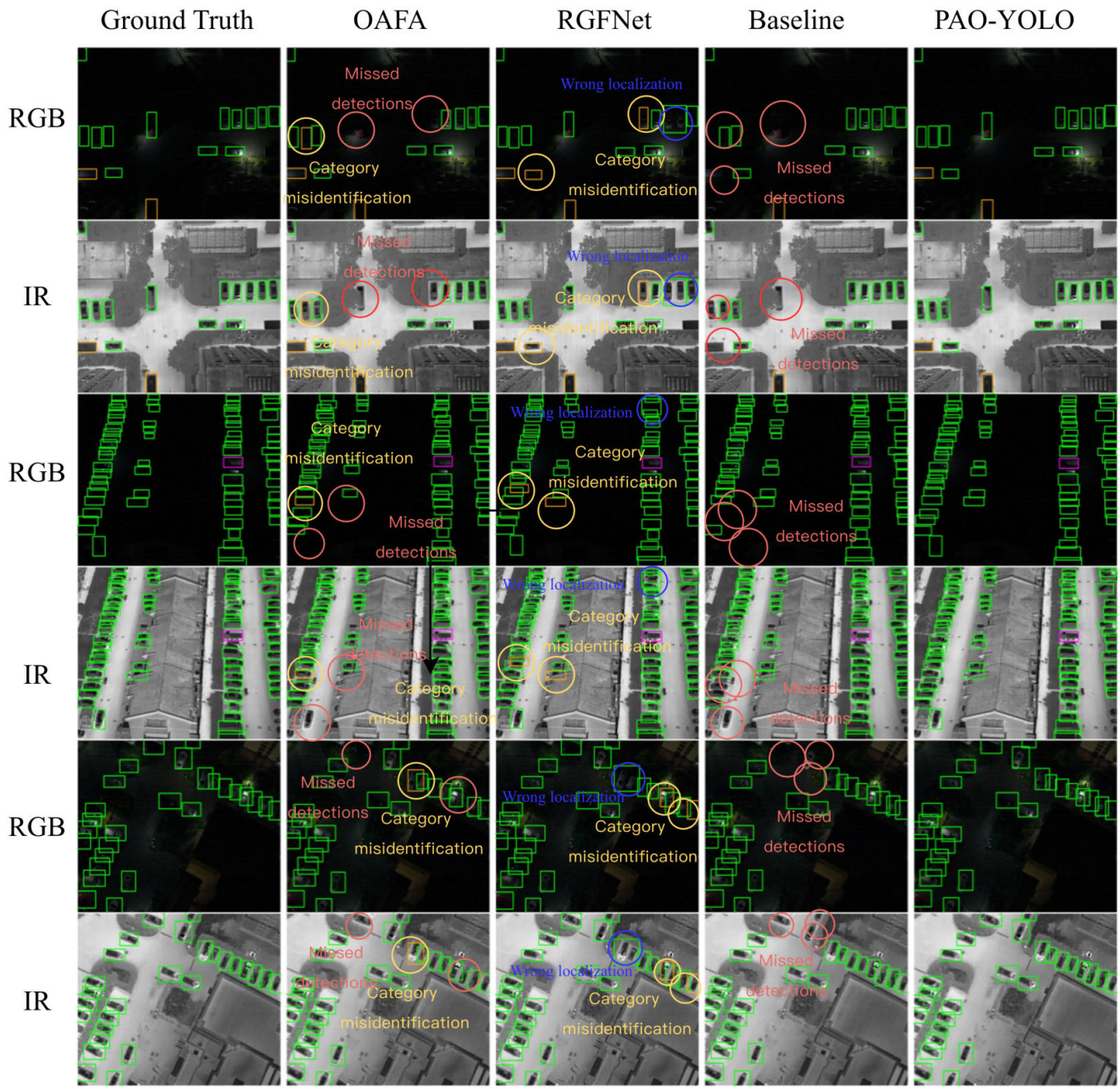


Fig. 7. Qualitative evaluation and interpretability analysis on the DroneVehicle and DVTOD datasets. (Top) Detection results where solid green boxes denote successful detections (true positives), while red, yellow, and blue dashed circles/boxes indicate missed detections (false negatives), wrong classification, and wrong localization, respectively. (Bottom) Heatmap visualization depicting the evolution of feature activation through the DSB modules. PAO-YOLO facilitates superior semantic consistency and localization precision by filtering modality-specific clutter.

memory thrashing. As a result, the full model sustains an inference speed of 85.60 FPS. This speed maintains a substantial margin over the standard 30 FPS real-time threshold, confirming that PAO-YOLO effectively resolves cross-modal semantic conflicts while preserving the real-time processing capability required for UAV operations.

F. Evaluation on the DVTOD Dataset

The DVTOD dataset serves as a rigorous benchmark for evaluating registration robustness, as it exhibits substantial spatial biases and nonrigid physical deviations between sensors. Detecting targets in such environments requires robust spatial compensation to prevent feature misalignment. The EDAF module in PAO-YOLO facilitates pixel-level elastic correction by computing prior-guided offset fields $\boldsymbol{\phi}_{\text{total}}$, effectively mitigating the impact of dynamic parallax. As summarized in Table IV, PAO-YOLO achieves an mAP@0.5 of 89.2%, yielding a quantifiable gain over the reflectance-guided RGFNet [32]. Notably, our framework maintains high precision across all categories (Person, Car, and Bicycle),

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demonstrating that explicit parallax compensation is more effective than simple feature addition for handling high-dynamic UAV imagery.

G. Qualitative Analysis and Interpretability

The visual performance and representational mechanism of PAO-YOLO are illustrated in Fig. 7. Quantitative detection comparisons (top) indicate that baseline detectors yield a high miss rate (red circles) under nocturnal conditions, primarily due to the low signal-to-noise ratio (SNR) in the VL channel. Furthermore, existing multimodal frameworks [32], [34] often exhibit category-level ambiguity (yellow circles) when infrared signatures lack fine-grained textural detail. In contrast, PAO-YOLO exploits TIR-based geometric constraints to compensate for corrupted VL regions. By establishing precise pixel-level alignment within physical coordinates, the framework effectively suppresses registration-induced artifacts. The interpretability analysis via activation maps (bottom) further characterizes the internal feature refinement. Compared to raw low-level features, the representations processed through the MIFD and EDAF modules exhibit a concentrated response on foreground manifolds while attenuating background clutter. This evolution of feature saliency indicates that the DSB mechanism effectively filters modality-specific noise and facilitates cross-modal consensus [13]. Such gradient purification ensures robust perception for sub-32-pixel targets in densely populated remote sensing scenarios [11].

V. Conclusion

This article introduced PAO-YOLO to tackle the inherent challenges of multimodal object detection on UAVs, namely physical parallax, modality competition, and information loss for small targets. The core innovation lies in the synergy between the MIFD and EDAF modules, which achieve mechanism-driven feature disentanglement and pixel-level elastic alignment to eliminate parallax-induced artifacts [9], [30]. By ensuring spatial-semantic consistency before feature aggregation, our method effectively resolves representational competition between modalities [13]. The inclusion of a high-resolution P2 layer further preserves crucial textural gradients, boosting sensitivity to sub-32-pixel objects [11]. On the DroneVehicle and DVTOD datasets, PAO-YOLO sets a new SOTA with mAP@0.5 scores of 82.2% and 89.2%, validating its effectiveness in complex remote sensing environments.

Despite the demonstrated advantages, this research has certain limitations that warrant further investigation. First, while PAO-YOLO establishes a favorable tradeoff between accuracy and speed on the DroneVehicle and DVTOD benchmarks, its scalability across more varied, ultralarge-scale multimodal datasets requires further validation. Second, the explicit physical parallax prior heavily relies on the assumption of extracting relatively stable geometric structures. In extremely dynamic flight environments (e.g., severe turbulent air or rapid UAV movements), the resulting severe motion blur and abrupt viewpoint shifts may exceed the compensatory capacity of the current elastic alignment module. Similarly, in extreme weather conditions where both modalities suffer from simultaneous signal attenuation, the initial offset estimation may degrade. Third, although the framework validates its real-time inference capability on high-performance GPUs, the significant complexity introduced by the DSB paradigm and the irregular memory access patterns associated with the EDAF alignment procedures pose nontrivial deployment challenges on ultralow-power, resource-constrained edge processors typical of UAV platforms. Consequently, future research will focus on exploring hardware-aware model compression techniques, operator fusion strategies, and post-training quantization to guarantee sustained ultralow latency in strictly SWaP-constrained environments.

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